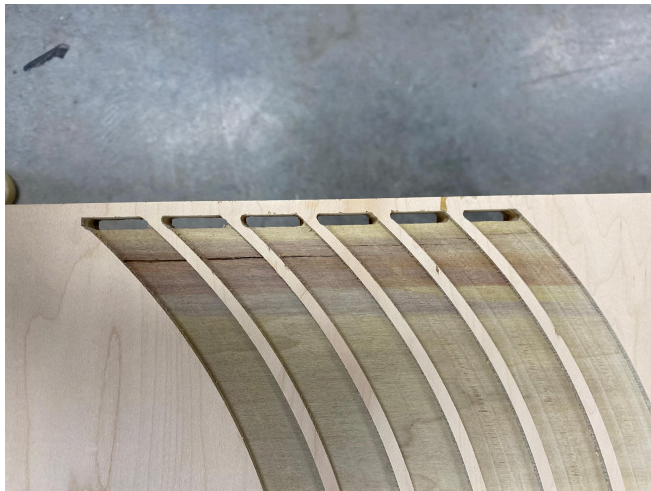


Element G

Prototype Explanation:



Step 1: The group began the construction process by importing the onshape document into a DXF file, and then cutting the wave design into a billet utilizing a CNC machine. The group then used portable orbital sanders to smoothen the surface and get rid of all bumps and chips.



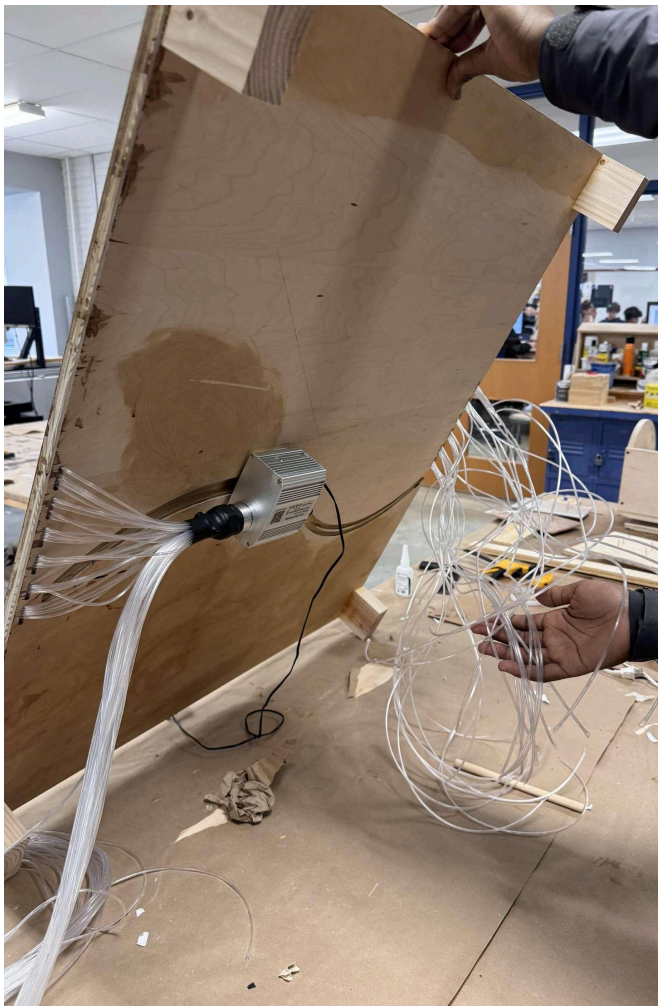
Step 2: The CNC machine failed to cut all the way through the slots, so the group utilized a portable drill to rout out the slots. The group then used files to sand the inside of the slots and remove any excess wood.



Step 3: The group then used Canary Wood wood stain to match the color of the wood in Whitemarsh's sensory room. This step took multiple attempts because it was the group's first time ever staining, so it was a learning experience. However, the group eventually achieved a satisfactory color.



Step 4: The group then painted the inside of the grooves black in order to make the color from the lights pop more. This conclusion was gathered after the group conducted research on which colors promoted the best reflectivity.



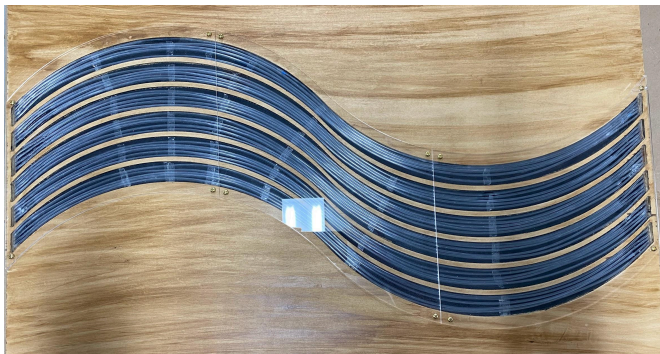
Step 5: The group then glued four 2x4 blocks to the back of the board, which acted as anchor points for when the board is screwed into the cubby. The group then utilized command strips to attach the fiber optic power source to the back of the board. All of the excess length left on the cables was cut to stay organized on the back.



Step 6: The group then utilized the laser engraver to make a plaque that was then glued down to the bottom left corner of the board. This plaque can be used to identify the group that was involved in designing and constructing the wave board.



Step 7: The group then utilized super glue and double sided tape to secure the cables to the inside of the grooves. There were five cables laid side by side per groove, and they entered/exited the grooves via the slots that the group drilled into the board.



Step 8: As the final step, the group utilized the laser engraver to cut out three acrylic sheets that were used to cover the fiber optic strands. These acrylics provide protection to the strands, but also ensure that they stay in place.

Final Prototype:



Testing or Modeling of Attributes:

Function	How it can be tested in our design
Change color	The final prototype has the ability to change colors through the use of a remote or a smartphone app. The group can test this function by discovering how many colors the board can display.
Fit into the sensory room	The final prototype has a set size, so we can test this function by measuring all dimensions of the board, and then ensuring that they fit within the constraints of the cubby in the sensory room. Additionally, this function can be tested by physically putting the prototype into the cubby and ensuring it lines up with the original design's measurements.
Control the brightness	The final prototype has the ability to change brightness through the use of a remote or a smartphone app. This function can be tested through a brightness measuring tool, with which the group can discover the range of the board's brightness

Have flashing lights/varying patterns	The final prototype has multiple premade patterns which vary in brightness, speed, and length. It also has the ability to create custom programs, which can be used to individually suit each student in the sensory room. This function can be tested by discovering how many different patterns the board can display.
Create light	The final prototype has the ability to light up. This function can be tested by measuring how many lumens the board is able to produce.
Have optional interactivity	The final prototype doesn't feature any interactive elements, but the primary stakeholder, Ms. Brennan, identified that interactivity was not of the utmost importance. However, the board can be touched and has space for potential future additions.

Addressing Non-Testable Attributes:

All elements of the design had the ability to be tested.